

HIROKI SAKAMOTO

Max Planck Institute for Dynamics of Complex Technical Systems, Germany
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RESEARCH INTERESTS

- Model reduction of high-dimensional dynamical systems
- Model compression for deep learning models
- Machine learning and data science for modeling and control of complex systems

AFFILIATIONS & APPOINTMENTS

Max Planck Institute for Dynamics of Complex Technical Systems 2026–present

Jun. 2026 – present Research Fellow

The University of Tokyo (UT) 2025–2026

Apr. 2026 – May 2026 JSPS Postdoctoral Fellow (PD)

Apr. 2025 – Mar. 2026 JSPS Research Fellow (DC2)

EDUCATION

The University of Tokyo Tokyo, Japan

Ph.D. in Information Science and Technology, Mar. 2026

Advisor: Kazuhiro Sato

M.S. in Information Science and Technology, Mar. 2023

Advisor: Kazuhiro Sato

Sorbonne University Paris, France

Exchange Student, Sep. 2023–Jan. 2024

Advisor: Emmanuel Trélat

Waseda University Tokyo, Japan

B.S. in Mathematics, Mar. 2021

Advisor: Hitoshi Arai

AWARDS

- Encouragement Award, ISCIE 2026
- Outstanding Presentation Award, The 68th Japan Joint Automatic Control Conference 2025
- Minister for Internal Affairs and Communications Award, Statistical Data Analysis Competition 2021

PUBLICATIONS AND PREPRINTS

- [1] H. Li, T. Nakata, **Hiroki Sakamoto**, and H. Uneya, “Lockdown policy rules with a hospital capacity constraint,” 2026, cARF Working Paper (CARF-F-620). [Online]. Available: <https://www.carf.e.u-tokyo.ac.jp/wp/wp-content/uploads/2026/03/F620.pdf>
- [2] **Hiroki Sakamoto** and K. Sato, “Data-driven regularized time-limited h^2 model reduction from noisy impulse responses,” *IEEE Control Systems Letters*, 2026.
- [3] **Hiroki Sakamoto** and K. Sato, “A deep state-space model compression method using upper bound on output error,” 2026, preprint. [Online]. Available: <https://arxiv.org/abs/2510.14542>

- [4] **Hiroki Sakamoto** and K. Sato, “Compression method for deep diagonal state-space model based on $\mathcal{H}^2$ optimal reduction,” *IEEE Control Systems Letters*, vol. 9, pp. 2043–2048, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/11087219>
- [5] **Hiroki Sakamoto** and K. Sato, “Data-driven h^2 model reduction for linear discrete-time systems,” 2024, preprint. [Online]. Available: <https://arxiv.org/abs/2401.05774>
- [6] M. Obara, K. Sato, **Hiroki Sakamoto**, T. Okuno, and A. Takeda, “Stable linear system identification with prior knowledge by riemannian sequential quadratic optimization,” *IEEE Transactions on Automatic Control*, vol. 69, no. 3, pp. 2060–2066, 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10258405>
- [7] **Hiroki Sakamoto** and K. Sato, “Stable probability of reduced matrix obtained by gaussian random projection,” *JSIAM Letters*, vol. 15, pp. 77–80, 2023. [Online]. Available: <https://doi.org/10.14495/jsiaml.15.77>
- [8] **Hiroki Sakamoto** and K. Sato, “Random projection preserves stability with high probability,” *JSIAM Letters*, vol. 15, pp. 17–20, 2023. [Online]. Available: <https://doi.org/10.14495/jsiaml.15.17>
- [9] **Hiroki Sakamoto** and K. Kawamoto, “Exploring factors driving young people’s migration from metropolitan to rural areas: An analysis using a modified gravity model,” *Tokei (Statistics)*, vol. 73, no. 3, pp. 56–63, 2022, (in Japanese).

CONFERENCE PRESENTATIONS

International Conferences

1. *Model compression for deep state space models using optimal- $\mathcal{H}^2$ model reduction*
Hiroki Sakamoto, Kazuhiro Sato, University of Manchester and University of Tokyo Joint Research Symposium: Global Green Transformation Workshop, University of Manchester, UK, Oct. 2024
2. *$\mathcal{H}^2$ Model Order Reduction for Deep State-Space Models*
Hiroki Sakamoto, Kazuhiro Sato, SICE Festival 2024 with Annual Conference, Kochi University of Technology, Japan, Aug. 2024

Conferences (Japan)

1. *Deep SSM Compression Without Retraining Using Output Error Upper Bounds: Evaluation on LRA benchmark and WikiText-103*
Hiroki Sakamoto, Kazuhiro Sato, The 13th SICE Multi-Symposium on Control Systems, Toyama International Conference Center, Japan, Mar. 2026
2. *A Deep State-Space Model Compression Method with Guaranteed Upper Bounds on Output Error*
Hiroki Sakamoto, Kazuhiro Sato, The 68th Japan Joint Automatic Control Conference, Nagoya University, Japan, Nov. 2025
3. *Compression of deep state-space models using the $\mathcal{H}^2$ optimal model reduction*
Hiroki Sakamoto, Kazuhiro Sato, The 12th SICE Multi-Symposium on Control Systems, Osaka Institute of Technology, Japan, Mar. 2025

4. *Data-driven h^2 model reduction for linear discrete-time systems*
Hiroki Sakamoto, Kazuhiro Sato, The 11th SICE Multi-Symposium on Control Systems, Hiroshima University, Japan, Mar. 2024
5. *Fast solving of generalized Sylvester matrix equations $AX + YB = C$*
Hiroki Sakamoto, Pierre-Henri Tournier, Emmanuel Trélat, Symposium on Cone Linear Programming and Related Topics, Seikei University, Japan, Feb. 2024
6. *Estimation of absolute constants in concentration inequalities based on Gaussian embeddings*
Hiroki Sakamoto, Kazuhiro Sato, The 19th JSIAM Activity Group Joint Workshop, Okayama University of Science, Japan, Mar. 2023
7. *Model Reduction of Large-Scale Stable Linear Systems via Random Projection*
Hiroki Sakamoto, Kazuhiro Sato, JSIAM Annual Meeting 2022, Online, Sep. 2022
8. *Analysis of factors influencing youth migration from metropolitan areas to rural regions using a gravity model*
 Kota Kawamoto, **Hiroki Sakamoto**, Japan Society of Economic Statistics 2022 Annual Meeting, Online, Sep. 2022

Other Talks

1. *Compression Method for Deep Diagonal State Space Model Based on $\mathcal{H}^2$ Optimal Reduction*
Hiroki Sakamoto, Kazuhiro Sato, Young Researchers' Camp on Control Theory 2025, Hiroshima, Japan, Sep. 2025
2. *Compression Method for Deep Diagonal State Space Model Based on $\mathcal{H}^2$ Optimal Reduction*
Hiroki Sakamoto, Kazuhiro Sato, CSC Seminar, Max Planck Institute for Dynamics of Complex Technical Systems, Magdeburg, Germany, Aug. 2025
3. *Parameter reduction of deep learning models using insights from control theory*
Hiroki Sakamoto, Kazuhiro Sato, Informatics for All by All, The University of Tokyo, Japan, Mar. 2025
4. *Fast solving of generalized Sylvester matrix equations $AX + YB = C$*
Hiroki Sakamoto, Pierre-Henri Tournier, Emmanuel Trélat, Workshop (Title TBD), University of Bologna, Italy, Dec. 2023
5. *Integrated model of infection and the economy: Literature review*
 H. Uneya, S. Kawawaki, **Hiroki Sakamoto**, K. Haganuma, The 1st COVID-19 Policy Study Meeting, Nagoya, Japan, Sep. 2022

GRANTS AND FELLOWSHIPS

JSPS Postdoctoral Fellowship (PD), JSPS	Apr. 2026 – May 2026
JSPS Research Fellowship (DC2), JSPS	Apr. 2025 – Mar. 2026
Overseas Study Support Program, JASSO	Sep. 2023 – Jan. 2024
Fellowship for Creation of Intelligent World, IIW, UT	Apr. 2023 – Mar. 2026
Grant for Independent Research, IIW, UT	Aug. 2022 – Mar. 2023

RESEARCH ASSISTANTSHIPS AND INTERNSHIP

Collaborative Project with Industry

Aug. 2024 – Mar. 2025

Research and development of deep-learning models for industrial systems.

Graduate School of Economics, UT

Feb. 2022 – Feb. 2024

Development of a Healthcare Forecasting Tool for Japan using the SIR Model. See [the website](#).

Recruit Co., Ltd.

Sep. 2022 – Dec. 2022

Improved the point-awarding algorithm for Hot Pepper.

TEACHING EXPERIENCE

Teaching Assistant, UT

- Mathematical Optimization, Oct. 2022 – Feb. 2023
- C Language Programming Exercises, Apr. 2021 – Aug. 2021

Teaching Assistant, Waseda University

- Fundamentals of statistics, ML, and programming, Apr. 2020 – Mar. 2021

TECHNICAL SKILLS

- Programming languages: MATLAB, Python.
- Software: PyTorch, TensorFlow, Google Cloud Platform.